

ENVS 193DS Final

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Link to GitHub repo [here](#).

Set up

```
# loaded in packages
library(tidyverse)
library(janitor)
library(here)
library(ggplot2)
library(readxl)
library(dplyr)
library(effsize)
library(MuMIn)
library(DHARMA)
library(ggeffects)

# read in data and saved as object called `nest`
nest <- readr::read_csv(here::here("data","nest_data_final.csv"))

# read in data and saved as object called `rxntime`
rxntime <- readr::read_csv(here::here("data","rxntime.csv"))
```

Problem 1. Research writing

a. Transparent statistical methods

In part 1, they used logistic regression. In part 2, they used a chi-square test of independence.

b. Suggestions for rewriting

Part 1. We found that the distance of microhabitat to water edge predicts the likelihood of usage by American Avocets.

Part 2. We found that different bird species (American Avocet, Forster's Tern, Black-necked Stilt) show preference for certain habitat types (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat).

c. Further analysis

Predation pressure can be studied as a discrete variable (count of individuals) or as a categorical variable (levels of predator population density); the presence of predators could make microhabitats too risky for the birds to frequent safely. As such, the birds may choose to avoid predator territories, reducing the probability of American Avocet microhabitat use in those locations. Predator data may be acquired through methods such as tracking, video sightings/identification, and/ or direct observation.

Problem 2. Data analysis

a. Clean and wrangle

```
# created new object from nest data frame
nests_clean <- nest |>
  # selected columns of interest
  select(height_cm, ant_species, case_control) |>
  # filtered ant species
  filter(
    ant_species == "AB" |
    ant_species == "RRB" |
    ant_species == "BBR") |>
  # reordered the levels of ant species
  mutate(ant_species = factor(ant_species,
                              levels = c("AB",
                                           "RRB",
                                           "BBR"))) |>
  # created a new column to store ant scientific names
  mutate(ant_species = case_when(
    ant_species == "AB" ~ "Crematogaster sjostedti",
    ant_species == "RRB" ~ "Crematogaster mimosae",
    ant_species == "BBR" ~ "Crematogaster nigriceps"))

# converted ant species to factor
nests_clean$ant_species <- factor(
```

```

nests_clean$ant_species,
levels = c("Crematogaster sjostedti",
           "Crematogaster mimosae",
           "Crematogaster nigriceps"))

# display of data in a sliced sample
slice_sample(nests_clean,
             n = 10,
             by = NULL,
             weight_by = NULL,
             replace = FALSE)

```

```

# A tibble: 10 x 3
  height_cm ant_species case_control
  <dbl> <fct>          <dbl>
1    318. Crematogaster nigriceps      1
2    174. Crematogaster mimosae      1
3    141  Crematogaster nigriceps      0
4    228  Crematogaster mimosae      1
5    168. Crematogaster mimosae      0
6    118  Crematogaster mimosae      0
7    212. Crematogaster mimosae      1
8    163  Crematogaster mimosae      0
9    389  Crematogaster mimosae      1
10   463  Crematogaster sjostedti      0

```

b. Response variable

A tree that contained a nest (“used” tree) is denoted with a (1) and trees without nests (“available” trees) are denoted with a (0).

c. Purpose of study

Tree height is a continuous variable, whereas ant species is a categorical variable. Trees that are taller may be older and harder to reach, offering more security for birds avoiding disturbance caused by “catastrophic (lethal) herbivory” and in turn, increasing the probability of probability of nest occurrence (Introduction, paragraph 2). The probability of probability of nest occurrence may differ depending on the aggressive tendencies (or lack thereof) of the ant species that has colonized the tree. More aggression to protect the tree may benefit birds’ nests from herbivory disturbance, but it may also put the birds and their offspring at risk of attack by hostile ants (Introduction, paragraph 3).

d. Table of models

Model number	Tree height	Ant species	Model description
1	x	x	model with interaction term
2	x	x	model w/o interaction term

e. Run the models

f. Select the best model

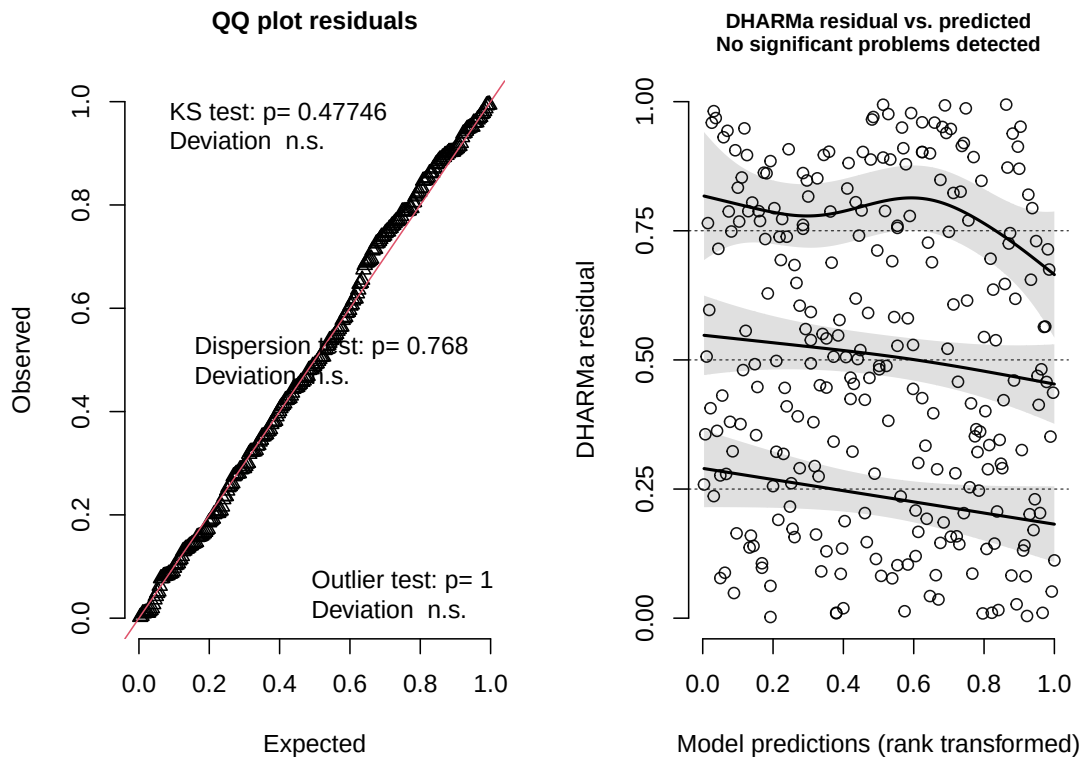
```
# used AIC to choose the best model
# made output invisible
invisible(AICc(
  model1, # interaction
  model2 # no interaction
))
```

The best model as determined by Akaike's Information Criterion (AIC) is the one that includes tree height and water treatment (with interaction terms) as predictors.

g. Check the diagnostics

```
# simulated residuals
plot(
  simulateResiduals(model1)
)
```

DHARMA residual



h. Visualize the model predictions

```
# stored custom colors
my_cols <- c(
  "Crematogaster sjostedti" = "hotpink",
  "Crematogaster mimosae"   = "gold3",
  "Crematogaster nigriceps" = "cyan2"
)

# predictions by ant species and tree height
mod_preds <- ggpredict(model1, terms = c("height_cm [all]",
                                         "ant_species")) |>
  # matched model predictions and geom_point column names
  rename(ant_species = group)
```

```
# aligned factor levels with raw data
mod_preds$ant_species <- factor(mod_preds$ant_species,
                                levels =
                                levels(nests_clean$ant_species))

# matched my_cols to factor levels
names(my_cols) <- levels(nests_clean$ant_species)
```

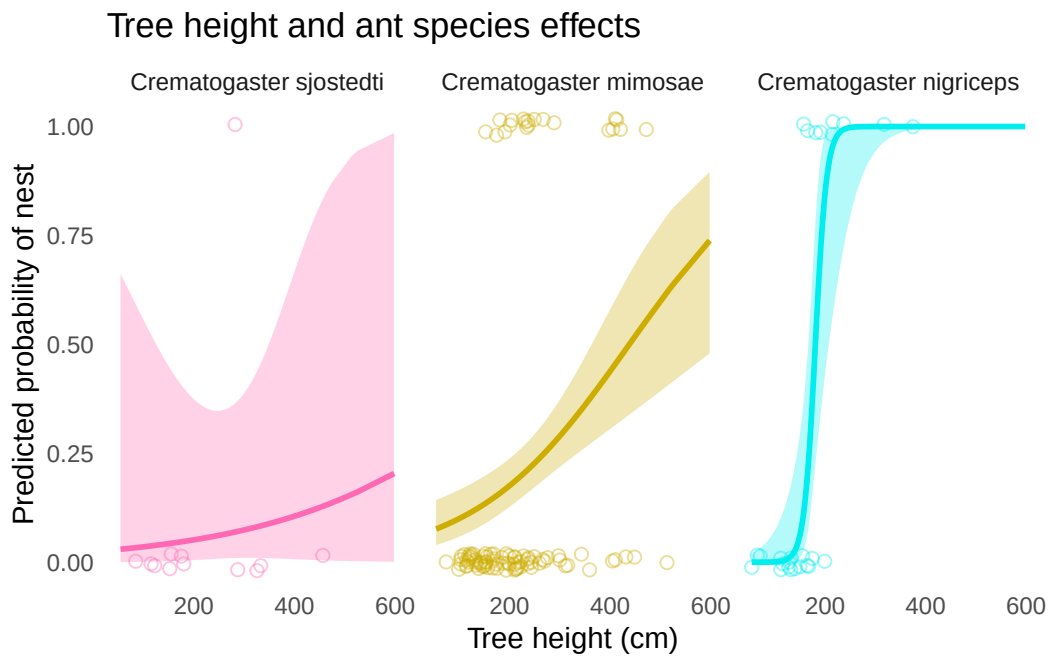
```
# created new object from nests_clean
nests_vis <- nests_clean |>
  # filtered case_control columns
  filter(!is.na(case_control) & case_control == c(0,1))

# base layer: ggplot using clean data frame
ggplot(nests_vis,
       # x-axis: tree height
       # y-axis: predicted probability of nest
       aes(x = height_cm,
           y = case_control)) +
  # first layer: raw data points
  geom_jitter(aes(color = ant_species), # color by ant species
             height = 0.02, # jitter height
             alpha = 0.4, # point opacity
             size = 2, # point size
             shape = 21) + # point shape
  # second layer: ribbon from predictions
  geom_ribbon(data = mod_preds, # model predictions data
            aes(x = x, # model predictions x-axis
                ymin = conf.low, # confidence low
                ymax = conf.high, # confidence high
                fill = ant_species), # color by ant species
            inherit.aes = FALSE, # reject default aesthetics
            alpha = 0.3) + # ribbon opacity
  # third layer: line from predictions
  geom_line(data = mod_preds, # model predictions data
           aes(x = x, # model predictions x-axis
               y = predicted, # model predictions y-axis
               color = ant_species), # color by ant species
           inherit.aes = FALSE, # reject default aesthetics
           linewidth = 1) + # line thickness
  facet_wrap(~ ant_species) + # facet by the same variable
  # used custom colors
```

```

scale_color_manual(values = my_cols) +
# filled with custom colors
scale_fill_manual(values = my_cols) +
# limited y-axis range to 0-1
coord_cartesian(ylim = c(0,1)) +
labs(x = "Tree height (cm)", # relabeled x-axis
     y = "Predicted probability of nest",
     title = "Tree height and ant species effects") + # added title
# custom theme
theme_minimal() +
# removed legend
theme(legend.position = "none", # removed legend
      panel.background = element_blank(), # removed panel background
      panel.grid = element_blank(), # removed panel grid
)

```



i. Write a caption for your figure

Figure 1. Predicted probability of nest occurrence increases with tree height across ant species. Open circles represent individual tree observations. Predicted probabilities of nests are graded on a scale of 0-1, with 0 indicating no nest and 1 indicating the occurrence of a nest, although points are jittered for visual clarity. Panels depict different ant species

that colonize the trees (pink: *C. sjostedti*, yellow: *C. mimosae*, blue: *C. nigriceps*). Ribbon depicts the upper and lower bounds of the confidence interval, and line represents the predicted probability of nest occurrence Data source: Lujan, E.; Nielsen, R.; Short, Z.; Wicks, S.; Nderitu Watetu, W.; Malingati Khasoha, L.; Palmer, T. M.; Goheen, J. R.; & Alston, J. M. (2023). Data from: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>

j. Calculate model predictions

```
# created an object for the tallest tree in the dataset
height_600 <- ggpredict(model1,
                        terms = c("height_cm [600]", # occurrence at 600 cm
                                "ant_species")) # probability by ant species

# displayed data
height_600
```

Predicted probabilities of case_control

ant_species: *Crematogaster sjostedti*

height_cm	Predicted	95% CI
600	0.20	0.00, 0.99

ant_species: *Crematogaster mimosae*

height_cm	Predicted	95% CI
600	0.74	0.48, 0.90

ant_species: *Crematogaster nigriceps*

height_cm	Predicted	95% CI
600	1.00	1.00, 1.00

k. Interpret your results

Tree height and ant species interact to predict probability of probability of bird nest occurrence in African savanna. At a tree height of 600 cm, predicted nest occurrence probability is 1.00 (95% CI: [1.00,1.00]) for *C. nigriceps*, 0.74 (95% CI: [0.48,0.90]) for *C. mimosae*, and 0.20 (95% CI: [0.00, 0.99]) for *C. sjostedti*. These findings are reflected in Figure 2, where trees colonized with *C. nigriceps* have the highest predicted nest occurrences and trees colonized with *C. sjostedti* have the lowest predicted nest occurrences.

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

In the affective visualization from Homework 3, each observation is represented by an icon, resulting in an array of 30 icons. The exploratory visualizations I made in Homework 2 were plotted so that the response variable (mean reaction time in milliseconds) would be on the y-axis and the predictor variables (exercise duration in minutes for the scatter plot and exercise type for the strip plot) would be on the x-axis. Observations were denoted with points (in the strip plot, points were assigned unique colors by exercise type).

Between all my visualizations, a common feature they share is a round, circular representation for observations. In the exploratory visualizations, these are the plotted points, while the affective visualization is made up of round volleyball and bike tire icons. Other than visual elements, the plots share the same response variable (mean reaction time in milliseconds). The strip plot and affective visualization both present information framed by the categories of exercise type.

Starting with exploratory visualizations, the strip plot displays a wide range of response values for biking and a more narrow range for volleyball (due to the lower observation count). The data collection incomplete at the time, it was hard to tell if there was a difference in central tendency for mean reaction time between the groups. The scatter plot was similarly difficult to draw concrete statements from due to low observation counts. Most points were concentrated in shorter durations around 7 minutes because I'd collected more data focused on biking (and biking took less time than volleyball sessions). Taking this into account, the negative trend shown in the plot can be understood as a very vague pattern. And in the affective visualization, volleyball and biking appeared to be the most frequent activities based on the number of icons, while volleyball activities involved the most social interaction. Patterns are different between visualizations because I had attempted to highlight a different finding through each figure.

The feedback I received during week 9 in workshop were mainly focused on making an artistic (affective) visualization because at the time I did not have one. I was also encouraged to record more data. My group's comments motivated me to create a visualization that was easy to understand and that would clearly communicate my findings. With their suggestions in

mind, I went with an array of icons that would signal different factors such as socialization and hydration with additional colors and smaller icons.

b. Designing an analysis

Based on my response variable (mean reaction time in milliseconds) and chosen predictor value (exercise type), a Mann-Whitney U test can be used to analyze my data in addition to answering my question: Does my mean reaction time differ across different kinds of exercise? This analysis will compare the central tendencies of a continuous variable (mean reaction time) between two categorical variables (exercise type).

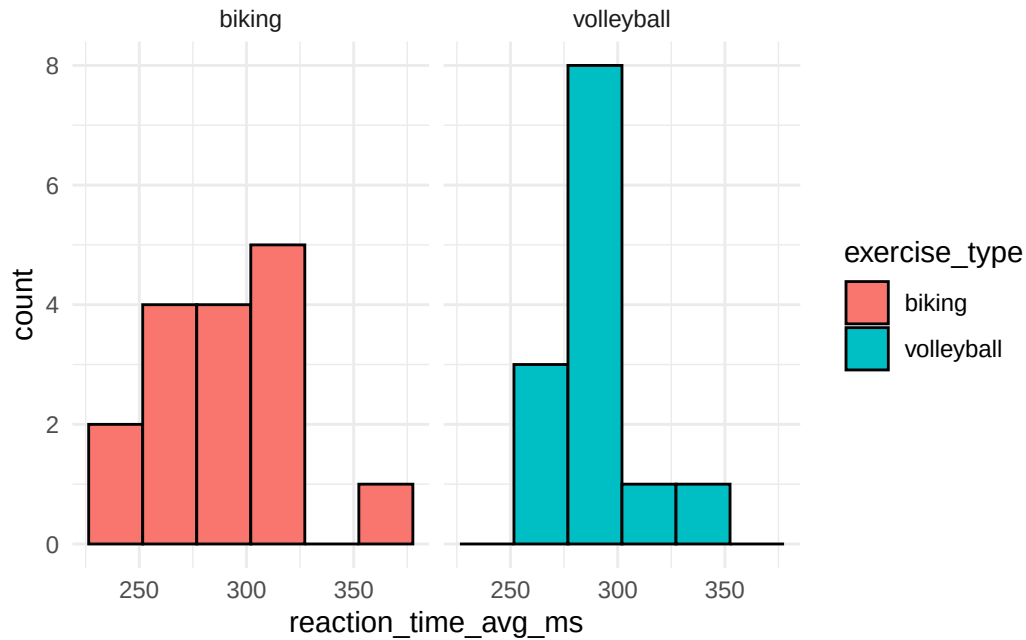
c. Check any assumptions and run your analysis

Cleaning data

```
# created new object from rxntime data frame
rxntime_clean <- rxntime |>
  # cleaned names
  clean_names() |>
  # filtered exercise types to exclude walking (insufficient data)
  filter(exercise_type != "walking")
```

Checking assumptions - Assessing normality with histograms

```
# base layer: ggplot using clean data frame
ggplot(data = rxntime_clean,
  # x-axis: mean reaction time
  # fill color by exercise type
  mapping = aes(x = reaction_time_avg_ms,
    fill = exercise_type)) +
  # first layer: histogram
  geom_histogram(bins = 6, # according to Rice Rule
    color = "black") + # outline
  facet_wrap(~ exercise_type) + # separate plots by exercise type
  # non-default theme
  theme_minimal()
```



Distributions are similarly right-skewed and data were randomly sampled, so a Mann-Whitney U test is appropriate for this data.

Running the analysis

```
# Mann-Whitney U test using clean data frame
wilcox.test(
  reaction_time_avg_ms ~ exercise_type,
  data = rxntime_clean
)
```

Wilcoxon rank sum test with continuity correction

```
data: reaction_time_avg_ms by exercise_type
W = 109.5, p-value = 0.8264
alternative hypothesis: true location shift is not equal to 0
```

Calculating effect size

```
# calculated effect size using clean data frame
cliff.delta(
  reaction_time_avg_ms ~ exercise_type,
  data = rxntime_clean
)
```

Cliff's Delta

```
delta estimate: 0.05288462 (negligible)
95 percent confidence interval:
      lower      upper
-0.3824587  0.4690131
```

d. Create a visualization

Cleaning data

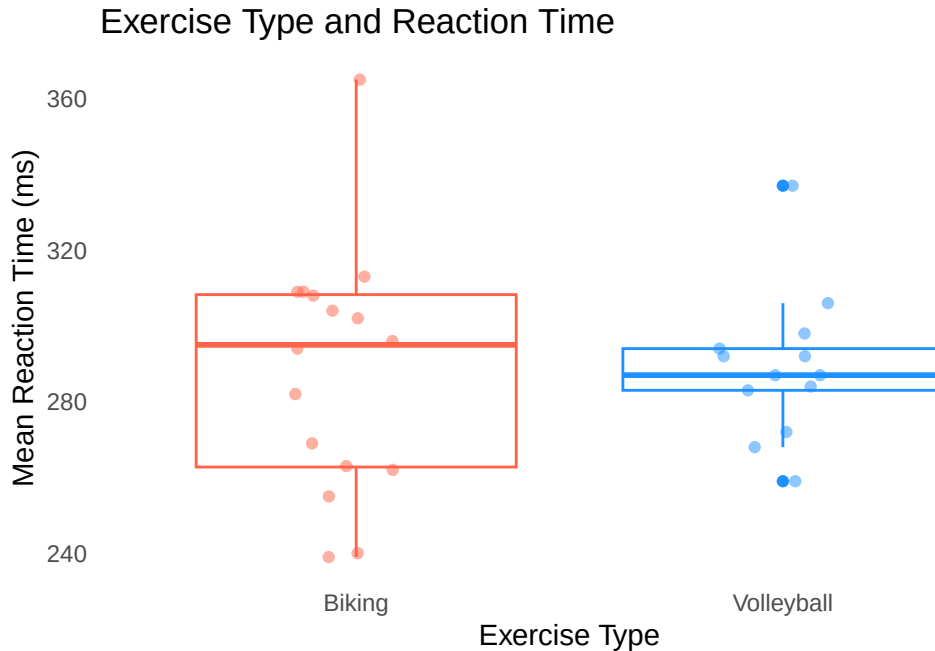
```
# created a new object from rxntime
rxntime_box <- rxntime |>
# cleaned column names
clean_names() |>
filter(exercise_type != "walking") |>
# selected columns of interest
select(exercise_type, reaction_time_avg_ms) |>
# capitalized existing abbreviations
mutate(exercise_type = case_when(
  exercise_type == "volleyball" ~ "Volleyball",
  exercise_type == "biking"     ~ "Biking")) |>
# removed missing values
drop_na()
```

Visualization

```

# base layer: ggplot
ggplot(data = rxn_time_box,
      # x-axis: exercise type
      # y-axis: mean reaction time
      # color by exercise type
      mapping = aes(x = exercise_type,
                    y = reaction_time_avg_ms,
                    color = exercise_type)) +
# first layer: boxplot
geom_boxplot() +
# second layer: data points and jitter
geom_jitter(width = 0.15, # range of horizontal jitter
            height = 0,    # no vertical jitter
            alpha = 0.5) + # lowered opacity of raw data
# relabeled x and y axes, added a title
labs(title = "Exercise Type and Reaction Time",
      x = "Exercise Type",
      y = "Mean Reaction Time (ms)") +
# custom theme
theme_minimal() +
# removed legend
theme(legend.position = "none") +
# custom colors
scale_color_manual(values = c("Biking" = "tomato",
                              "Volleyball" = "dodgerblue")) +
# removed grid and axis lines
theme(panel.grid = element_blank(),
      axis.line = element_blank())

```



e. Write a caption

Figure 2. Mean reaction times across three exercise types. Box/ jitter plots of mean reaction times (ms) for three exercise types (red: biking, blue: volleyball). Interquartile ranges fall around 260-310 milliseconds and median values are 295 and 287 milliseconds for biking and volleyball, respectively. Points represent individual observations. Data are from my personal observations throughout the spring quarter.

f. Write about your results

We found no statistically significant difference in median reaction times between volleyball (median = 287 ms, $n = 13$) and biking (median = 295 ms, $n = 16$) exercises (Wilcoxon rank-sum test, $W = 109.5$, $p = 0.83$, $\alpha = 0.05$). There was a small (Cliff's $\delta = 0.05$) difference in median reaction times between exercise types.

g. Sharing your affective visualization

I was there that day
 Lovely visualizations
 Everywhere, outside